DESIGN

Helix OTA — Emulation/Virtualization Test-Fabric — Architecture & Design

Field	Value
Revision	2
Last modified	2026-06-11T10:30:00Z
Status	active — T0 floor + Tab dev-host A/B-virt FOUNDATION built; Tab slot mechanism in progress
Status summary	The incorporation design for a LOCAL + DISTRIBUTED hardware-free test fabric, derived from the cited verdicts in `../../research/emulation_infra/REPORT.md` and the dev-host A/B-virt work in `../../research/rk3588_emulator/REPORT.md`. Extends the `containers` submodule (§11.4.76) and the existing Tier-1/2/3 baseline in `../EMULATED_DEVICE_TESTING.md`. Reconciles the macOS-M3 dev host vs the Linux-KVM CI tier as FACT (§11.4.6/ §11.4.112). **Now partially built:** the T0 floor (shipped) plus a new **Tab dev-host A/B-virt tier** (QEMU virt+HVF + U-Boot bootcount/ altbootcmd + RAUC dm-verity) whose FOUNDATION boots to live userspace (PROVEN); its real A/B slot switch / dm-verity / auto-rollback is **in progress**, gated on the in-flight `u-boot.bin` build — NOT proven (§11.4.6).
Authority	Operator mandate 2026-06-10 (hardware-free comprehensive emulation/VM/container fabric)
Related	`../../research/emulation_infra/REPORT.md`; `../../research/rk3588_emulator/REPORT.md`; TEST_COVERAGE_PLAN.md; SCHEMA.sql; ROADMAP.md; `containers/` submodule

1. Goal & non-goals

Goal. A single fabric that boots/runs emulated, virtual, and (where wired) real targets — locally on the dev host AND distributed across remote/CI nodes — so every Helix OTA work item is testable WITHOUT a physical RK3588/Orange Pi on the operator's desk. Anti-bluff is the spine: a tier never claims fidelity it does not have (§11.4 / §11.4.107), and the tier boundaries below are stated as FACT precisely so no tier silently over-claims.

Non-goals. (a) Reinventing container/VM machinery — every primitive is an EXTENSION of `vasic-digital/containers` (§11.4.74/§11.4.76), never a re-implementation. (b) Pretending the macOS-M3 host can run KVM-gated Android — it cannot (§3); that is an honest host gap (§11.4.112), not a thing to fake.

2. Tier model (reconciled with EMULATED_DEVICE_TESTING.md)

Tier	Fidelity	Tech (from REPORT verdicts)	Where it runs	What it proves
T0 — protocol client	wire/server flow only (no real apply)	existing `ota-device-emulator` (`server/internal/deviceemu`) on podman	**dev host + anywhere**	register→update-check→telemetry→delta→rollout→ against a live control plane
T1 — Android userspace	real Android app/agent logic; NOT real A/B apply	Android **AVD arm64-v8a** (HVF local / KVM CI); Genymotion optional	**dev host (HVF, accelerated)** + Linux/KVM CI	`ota-android-agent`/bridge decisio logic, ADB-driven UI/automation
Tab — dev-host A/B-virt (built; slot mechanism in progress)	**real U-Boot bootcount/altbootcmd A/B slot-switch + RAUC dm-verity + auto-rollback** on a generic aarch64 virt board (NOT the RK3588 SoC, NOT Android `update_engine`)	**QEMU `-machine virt` + HVF** aarch64 guest (Buildroot) + U-Boot + RAUC (`tests/emulator/ab_virt/`)	**dev host (HVF, accelerated)**	the real A/B slot-switch + rollback semantics on this Apple-Silicon host WITHOUT a KVM gate — closes pa the T0→T2 gap locally. **FOUNDAT boots to live userspace (PROVEN) slot switch itself is authored-not-ye run (§11.4.6, gated on the `u-boot.` build).**
T2 — high-fidelity A/B	**real `update_engine` A/B + AVB/dm-verity + auto-rollback** on virtual partitions	**Cuttlefish (cvd)**	**Linux+KVM CI only** (host-gated on M3)	the genuine Android OTA apply + rollback semantics short of real silico
T3 — real hardware	vendor HAL + real U-Boot slot-switch + dm-verity on real partitions + Mali GPU	**RK3588/Orange Pi 5 Max behind LAVA**; `redroid-rk3588`	**HIL lab**	the only env exercising real silicon; flashes governed by §11.4.133

Tier	Fidelity	Tech (from REPORT verdicts)	Where it runs	What it proves
		adjacent (SoC app/ GPU, no bootloader)		
Tfw — firmware/ Linux-target	aarch64 UEFI/U-Boot/Linux boot (generic virt board, not RK3588 SoC)	**QEMU `-machine virt` + edk2** (existing `pkg/vm`); **Renode** future	**dev host (TCG)** + CI	U-Boot slot logic, future Linux/RTO targets
Tcp — control-plane isolation	N/A (not a device)	**Firecracker / Kata / Cloud Hypervisor** microVMs; gVisor sandbox	Linux+KVM CI (gVisor needs no KVM)	scaling/DDoS/chaos/security of the C control plane under real isolation

Cross-cutting: **distributed fan-out** via **LAVA** (one job schema drives BOTH virtual T0/T1/T2 and physical T3) and/or **Nomad** (heterogeneous scheduler for a mixed container+QEMU+cvd fabric); **CI front door** via **our own** self-hosted mac+Linux runners (GitHub-hosted Apple-Silicon runners do NOT expose HVF/nested-virt — FACT, REPORT §5).

```
flowchart TB
  subgraph DEV["Dev host — Apple M3 / macOS (HVF, no /dev/kvm)"]
    T0d["T0 ota-device-emulator (podman)"]
    T1d["T1 Android AVD arm64-v8a (HVF)"]
    TABd["Tab A/B-virt: QEMU virt+HVF + U-Boot + RAUC (foundation
booted; slot switch in progress)"]
    FWd["Tfw QEMU virt + edk2 / Renode"]
  end
  subgraph CI["Distributed CI — Linux + KVM nodes"]
    T2["T2 Cuttlefish (cvd) — real A/B/AVB/dm-verity"]
    T0c["T0 emulator fan-out (podman)"]
    T1c["T1 AVD/KVM farm"]
    TCP["Tcp Firecracker/Kata microVMs — CP scaling/chaos"]
  end
  subgraph HIL["HIL lab — real silicon"]
    T3["T3 RK3588 / Orange Pi 5 Max"]
    RD["redroid-rk3588 (SoC app/GPU)"]
  end
  CP["Helix OTA control plane (server/)"]
  SCHED["Fabric scheduler — LAVA job schema + Nomad/K8s exec"]
  RUN["CI front door — self-hosted mac+Linux runners"]
  RUN --> SCHED
  SCHED --> DEV & CI & HIL
  T0d & T1d & TABd & T2 & T0c & T1c & T3 & RD -->|ota-protocol| CP
  SCHED -->|exclusive lease per target §11.4.119| T2 & T3
```

3. The host fact (FACT, §11.4.6 — drives every placement)

Dev host = **Apple M3 Pro / macOS 15.5 / arm64**, accel = **HVF**, **no `/dev/kvm`**, and **M3 lacks hardware nested-virt** (M4+/macOS-15 only) — so KVM-gated Android (Cuttlefish, redroid, K8s-STF emulator nodes, Firecracker/Kata/Cloud-Hypervisor) is **host-gated here, NOT structurally impossible** (runs on a Linux-KVM box / M4+ / GCE nested-virt). The fabric therefore places T2/Tcp on the **Linux+KVM CI tier**, and keeps T0 + T1(AVD-HVF) + Tfw(QEMU-TCG) as the **locally-runnable** set. This split is asserted, never wished away.

4. Extension of the `containers` submodule (§11.4.74 extend-don't-reimplement)

Reuse (already present): `pkg/boot` + `pkg/compose` + `pkg/health` (on-demand boot + readiness — the §11.4.76 on-demand-infra invariant the pgx integration test already uses), `pkg/emulator` (AVD x86_64/arm64 + HVF/KVM accel gating + AVD-lock/orphan-qemu reaping), `pkg/vm` (QEMU aarch64 `virt`+UEFI), `pkg/genymotion`.

Tab dev-host A/B-virt (built — `tests/emulator/ab_virt/`): the Tab tier reuses **U-Boot** (bootcount/altbootcmd slot-select) and **RAUC** (dm-verity A/B update client) upstream per §11.4.74 — neither reimplemented in-project — on top of `pkg/vm`'s QEMU aarch64 `virt`+HVF boot. The guest image (Buildroot base + kernel) is built inside a named podman aarch64 Linux container (this host is macOS); the 2-slot GPT disk assembler (`assemble_ab_disk.sh`) + U-Boot `boot.cmd`/env (`uboot_ab/`) are authored. The base image build (PWU-AB-0) and the live-userspace boot (PWU-AB-1 FOUNDATION) are DONE+PROVEN; the slot-switch disk is gated on the in-flight `u-boot.bin` build (§11.4.6 — in progress).

New primitives to ADD upstream (PR to `vasic-digital/containers`), never copied in-project:

- `pkg/cuttlefish` — `cvd` lifecycle wrapper (fetch/launch/stop, `aosp_cf_arm64_only_phone`, KVM-presence gate that SKIPs-with-reason on a non-KVM host).
- `pkg/lava` — LAVA REST/XML-RPC client: submit a job, poll, pull artifacts; one job schema for virtual + physical DUTs.
- `pkg/fabric` — the distributed scheduler façade + **target registry** (lease/release a target exclusively, §11.4.119) over a Nomad/K8s/LAVA backend.

Stays in helix_ota (OTA-domain logic, not generic plumbing): `server/internal/deviceemu` (the T0 emulator — it speaks `ota-protocol`), the per-tier on-device test drivers, and the bank/Challenge wiring.

5. On-demand boot + single-resource-owner contracts

- **On-demand boot (§11.4.76):** the test entry point boots its tier's infra via the submodule (`pkg/boot/compose/health/fabric`); operators never hand-start podman, an emulator, `cvd`, or a LAVA job. A tier with unmet host prereqs **SKIPs-with-reason** (§11.4.3), never fakes a PASS.

- **Single-resource-owner (§11.4.119):** every exclusive target (a `cvd` instance, a real board, an HDMI/serial line, a fixed port) is leased to **exactly one** driver stream at a time via `pkg/fabric`'s registry; other streams targeting it are read-only or queued. Concurrent drivers of one exclusive target produce cross-contaminated evidence = a §11.4 bluff, so the lease is mandatory.

6. Persistence

A test-fabric registry (targets, runs, results, evidence paths, distributed-node inventory) is defined in `SCHEMA.sql`. It is OPTIONAL for local single-node use (the T0 in-process/podman tiers need no DB) and REQUIRED once distributed fan-out is wired (so the scheduler can lease targets and the dashboard can show fleet-of-fabric state). It reuses the project's pgx/Postgres seam, not a new datastore.

7. Anti-bluff posture (the spine)

Each tier produces captured evidence under `docs/qa/<run-id>/` per `TEST_COVERAGE_PLAN.md`: T0 captures real request/response transcripts; T1 captures ADB/UI + `update_engine_client` state; **Tab** captures the QEMU+HVF boot console (`docs/qa/20260611T061626Z-ab-virt-boot/console.log`, `HELIX_USERSPACE_LIVE_OK` — the FOUNDATION proof) and, once its disk is runnable, the slot-state assertion (`findmnt /` + `cat /etc/slot_id` across a switch) + the auto-rollback trace; T2 captures real `update_engine` A/B-slot + AVB/dm-verity + auto-rollback evidence; T3 adds on-silicon apply/rollback. A tier claiming a PASS without exercising the real system at its fidelity is a §11.4 PASS-bluff. Honest gaps (§3, REPORT §8) are documented, never hidden behind a green line — the Tab slot switch is recorded as **NOT proven** (gated on `u-boot.bin`) and T2 stays **host-gated** (no KVM on this macOS host), never as faked greens.